what are different frameworks and work plans like PICO, SMART and other tools I could use to start, write and work on my thesis?

Fantastic question — this is exactly the kind of strategic thinking that makes a **thesis organized, efficient, and professional.**

Here's a clear overview of the most useful **frameworks and planning tools** for **each phase** of your thesis — from defining your question to writing and presenting it.

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1. Defining and Structuring Your Research Question

PICO / PICo / SPIDER / CIMO / PECO

These help you translate broad ideas into structured, answerable research questions.

Framework	Best For	Elements	Example Use
PICO	Comparative / experimental research	Population – Intervention – Comparison – Outcome	Comparing biobased vs plastic packaging
PICo	Qualitative or descriptive studies	Population – Interest – Context	Exploring perceptions of sustainable packaging
SPIDER	Qualitative systematic reviews	Sample – Phenomenon of Interest – Design – Evaluation – Research type	Reviewing consumer attitudes
CIMO	Design / evaluative studies	Context – Intervention – Mechanism – Outcome	Evaluating how material design impacts performance
PECO	Environmental or toxicology studies	Population – Exposure – Comparator – Outcome	Assessing impact of exposure to microplastics

For your thesis, **PICO** (as you're already using) + a bit of **CIMO** (design and evaluation of materials) is ideal.

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2. Setting Objectives and Deliverables

SMART Goals

Define realistic and measurable milestones for your 24-week project.

SMART	Meaning	Example for You
Specific	Clear goal	"Create a FAIR dataset of biobased packaging materials"
Measurable	Quantifiable progress	"Include ≥50 unique material entries with metadata"
Achievable	Realistic	"Based on literature and accessible databases"
Relevant	Connected to research aims	"Supports EU sustainable packaging goals"
Time-bound	Has deadlines	"Complete data collection by Week 12"

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3. Literature Review and Data Gathering

PRISMA

A systematic review framework — you already mentioned it!
Use it to ensure transparency in how you search, screen, and include papers.

Steps:

1. Define research question (via PICO)
2. Identify databases and search strings
3. Screen titles & abstracts (inclusion/exclusion criteria)
4. Extract data (build Excel or Python-based extraction template)
5. Visualize the process in a PRISMA flow diagram

🧮 *Pro tip:* Combine PRISMA with **Rayyan** (a free tool for literature screening).

🗂 4. Organizing Data and Metadata

FAIR Principles

Since your output is a dataset, structuring it according to **FAIR** is crucial.

FAIR Principle	Meaning	Example in Your Project
Findable	Unique ID, metadata, DOI	Assign DOIs or metadata fields
Accessible	Public or open repository	Upload to Zenodo or Wageningen Data Portal
Interoperable	Standardized formats	Use CSV + JSON + units (SI)
Reusable	Clear license, documentation	Add data dictionary and references

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5. Analysis and Comparison Frameworks

SWOT / Multi-Criteria Decision Analysis (MCDA) / Comparative Tables

These help interpret data and guide conclusions.

Framework	Use	Example
SWOT	Strengths, Weaknesses, Opportunities, Threats	Assess bioplastics vs conventional plastics
MCDA	Quantitative comparison	Rank materials based on mechanical, barrier, and sustainability scores
Radar / Spider Chart	Visual comparison	Visualize performance across multiple metrics

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✍ 6. Writing and Argumentation Frameworks

IMRaD Structure

Most theses and journal papers follow:

Introduction → **M**ethods → **R**esults → **A**nd → **D**iscussion

Each section maps to your work:

- **Introduction:** Background + gap + objectives (use PICO/SMART)
- **Methods:** PRISMA, data extraction, FAIR structuring
- **Results:** Dataset overview + comparative figures
- **Discussion:** Implications, limitations, next steps

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7. Project & Time Management

Gantt Chart / Kanban / 3-Phase Plan

To stay on schedule during your 24-week timeline:

Phase 1 (Weeks 1–8): Setup & Literature Review
Phase 2 (Weeks 9–16): Data Extraction & Curation
Phase 3 (Weeks 17–24): Analysis, Writing & Visualization

Tools:

- **Trello** or **Notion** → for tasks
- **Excel Gantt Chart** or **MS Project** → for time planning

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🔄 8. Reflection and Iteration Tools

Framework	Purpose	Use Example
Gibbs' Reflective Cycle	Learning & evaluation	Reflect weekly on what worked or not
Design Thinking Loop	Iterative improvement	Empathize → Define → Ideate → Prototype → Test (e.g., in dataset structure)

✅ Summary Table

Thesis Stage	Framework/Tool	Purpose
Define	PICO, SMART	Research focus & objectives
Review	PRISMA	Literature search & selection
Collect	FAIR	Dataset design & standardization
Analyze	SWOT, MCDA	Comparative assessment
Write	IMRaD	Report structure
Manage	Gantt, Kanban	Time & workflow tracking
Reflect	Gibbs, Design Thinking	Learning and improvement

Would you like me to make this into a **visual roadmap (diagram)** — like a flowchart showing how all these frameworks connect throughout your 24-week project?
It could be a great slide for your supervisor presentation.

are there more? and yes i would love a visual roadmap

Perfect — you're thinking exactly like a researcher who wants both **depth and structure**.

Let's go **one level deeper**: below you'll find
1 ☐ A broader list of **research, project, and reflection frameworks** (beyond the usual ones like PICO, SMART, PRISMA, FAIR, IMRaD, etc.), and
2 ☐ A description of the **visual roadmap** I'll make for you next — showing *how* they fit together in your 24-week thesis.

🧭 A. Extended Framework Toolbox for Thesis Projects

Here's a comprehensive menu of frameworks — grouped by what stage they support:

🧮 1. Research Question & Concept Development

Framework	Purpose	Typical Use
PICO / PICo / SPIDER / CIMO / PECO	Structure research questions	Define the scope and focus of the thesis
5W1H (What, Why, Who, Where, When, How)	Quick scoping tool	Brainstorming and narrowing focus
Logic Model	Link inputs → activities → outputs → outcomes	Useful if you want to visualize your process or theory of change
Theory of Change (ToC)	Map assumptions → actions → impacts	Good for sustainability- or policy-oriented theses

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2. Goal Setting & Planning

Framework	Purpose	Example Use
SMART / OKR (Objectives & Key Results)	Concrete goal setting	Thesis milestones and outcomes
Gantt Chart / Kanban Board	Time and workflow management	Weekly or phase-based planning
RACI Matrix (Responsible, Accountable, Consulted, Informed)	Role clarity	If working with a supervisor or industry partner

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3. Literature Review & Data Gathering

Framework	Purpose	Example Use
PRISMA	Systematic review transparency	Selecting and screening literature
SALSA Framework (Search, Appraisal, Synthesis, Analysis)	Broader alternative to PRISMA	For narrative or scoping reviews
Arksey & O'Malley Scoping Review Framework	Identify key concepts and gaps	For exploratory research on emerging topics
TAPUPAS (Transparency, Accuracy, Purposivity, Utility, Propriety, Accessibility, Specificity)	Data quality appraisal	Checking reliability of extracted data

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🗂 4. Data Structuring & Analysis

Framework	Purpose	Example Use
FAIR Principles	Data stewardship	Structuring your dataset
MCDA (Multi-Criteria Decision Analysis)	Comparative evaluation	Ranking material performance
SWOT / TOWS	Strategic comparison	Assessing material pros and cons
KJ Method (Affinity Diagramming)	Organizing themes	Grouping literature findings
Gap Analysis	Identify missing information	Framing dataset value and novelty

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✍ 5. Writing & Communication

Framework	Purpose	Example Use
IMRaD	Standard scientific	Writing the report or article

Framework	Purpose	Example Use
	structure	
SCQA (Situation, Complication, Question, Answer)	Clear storytelling	For introduction or presentation flow
MECE Principle (Mutually Exclusive, Collectively Exhaustive)	Logical structuring	Organizing sections without overlap
Pyramid Principle	Top-down communication	Presenting arguments in PowerPoints or reports

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6. Reflection, Learning & Improvement

Framework	Purpose	Example Use
Gibbs' Reflective Cycle	Personal learning reflection	Weekly thesis journal
Kolb's Experiential Learning Cycle	Learning from action	Iterating on dataset design
Design Thinking Loop	Iterative prototyping	Building data structures or dashboards
PDCA (Plan-Do-Check-Act)	Continuous improvement	Testing data curation workflow
Lean Research Principles	Ethical, efficient research	Especially relevant for sustainability topics